

MORIAH COLLEGE

Year 11

Mathematics Extension 1

Preliminary Examination September 2015

Time Allowed: 90 minutes

Reading Time: 5 minutes

Examiner: J.Cohen, L.Bornstein, G. Lang

General Instructions

- Write using blue or black pen.
- Board-approved calculators may be used.
- All necessary working should be shown in every question.
- Start each question on a separate sheet.

Name

MULTIPLE CHOICE GRID

5 marks

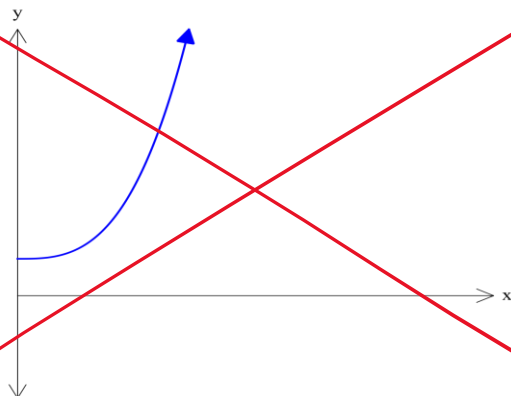
1.	A	B	C	D
2.	A	B	C	D
3.	A	B	C	D
4.	A	B	C	D
5.	A	B	C	D

Multiple choice (1 mark each)

1. Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 1}{2x^2 + x - 1}$.

- (A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) 1 (D) 0

2. The following graph has the following information relating to its $f'(x)$ and $f''(x)$



- (A) $f'(x) < 0$
 $f''(x) < 0$ (B) $f'(x) < 0$
 $f''(x) > 0$ (C) $f'(x) > 0$
 $f''(x) > 0$ (D) $f'(x) > 0$
 $f''(x) < 0$

3. Which term best describes the roots of the equation $x^2 + 4x - 3 = 0$?

- A) Real and different B) Equal
C) Definite D) Unreal

4. If $4x^2 + 3x - 5 \equiv Ax(x+1) + B(x+1) + C$, the value of B is:

- (A) 3 (B) -1 (C) 2 (D) -2

5. Which of the following divides the line segment from $A(4, -3)$ to $B(-8, 5)$ **externally** in the ratio 3:1?

- (A) $(-7, \frac{9}{2})$ (B) $(16, -9)$ (C) $(10, -7)$ (D) $(-14, 9)$

Question 6 (start each question on a new page) (15 marks)

a) Solve $\frac{4}{x+2} \geq \frac{1}{x}$ **3**

b) If one root of the equation $x^2 - mx + 2 = 0$ is double the other, find the value(s) of m. **2**

c) If $x = 3 - t$ and $y = 2t + 1$, find the Cartesian equation of the curve. **2**

d) If α, β are the roots of $2x^2 - 13x + 9 = 0$,

find the value of :

i) $(a+1)(b+1)$ **2**

ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ **2**

e) Solve the equation : $\tan^2 x - 3 \sec x + 3 = 0$ for $0 \leq x \leq 360^\circ$ **4**

Question 7 (start each question on a new page) (15 marks)

- a) For what range of values of k will the expression $kx^2 - (5k+1)x + 9k$ be positive definite. **3**

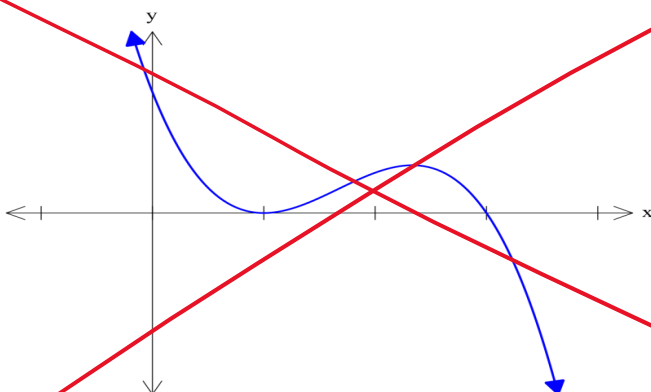
- b) A parabola has equation $y^2 = 8x + 2y + 7$.

- i) Express this equation in the form $(y - p)^2 = 4a(x - k)$. **1**

- ii) Draw a neat sketch of the parabola, clearly showing the vertex, focus and directrix. **4**

- iii) Find the endpoints of the latus rectum. **1**

- c) The above diagram shows a sketch of the curve $y = f(x)$.



On the attached page, draw a possible sketch of the function $y = f'(x)$ **2**

- d) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

- i) Derive the equations of the tangents at P and Q.

- ii) Show that the point of intersection of the tangents is

$$[a(p+q), apq]$$

4

Question 8 (start each question on a new page) (15 marks)

a) Consider the graph $y = \frac{x^2}{1-x^2}$

i) Find the stationary point(s) and their nature

3

ii) Find the equation of the horizontal and vertical asymptotes

2

iii) Sketch the graph.

2

~~b) Prove by mathematical induction that for all integers, $n \geq 1$ that~~

~~$$3 \cdot 2^2 + 3^2 \cdot 2^3 + 3^3 \cdot 2^4 + \dots + 3^n \cdot 2^{n+1} = \frac{12}{5}(6^n - 1)$$~~

4

c) Solve for x:

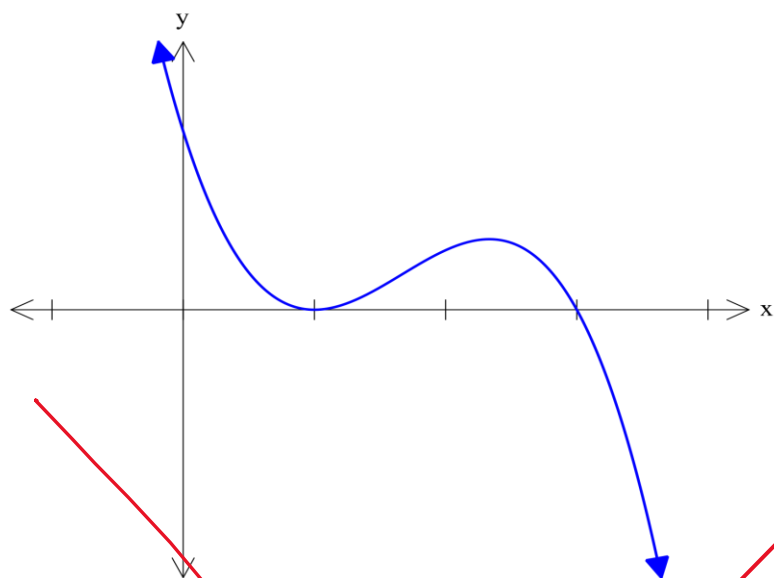
$$(x^2 + 2x)^2 = 14x^2 + 28x + 15$$

4

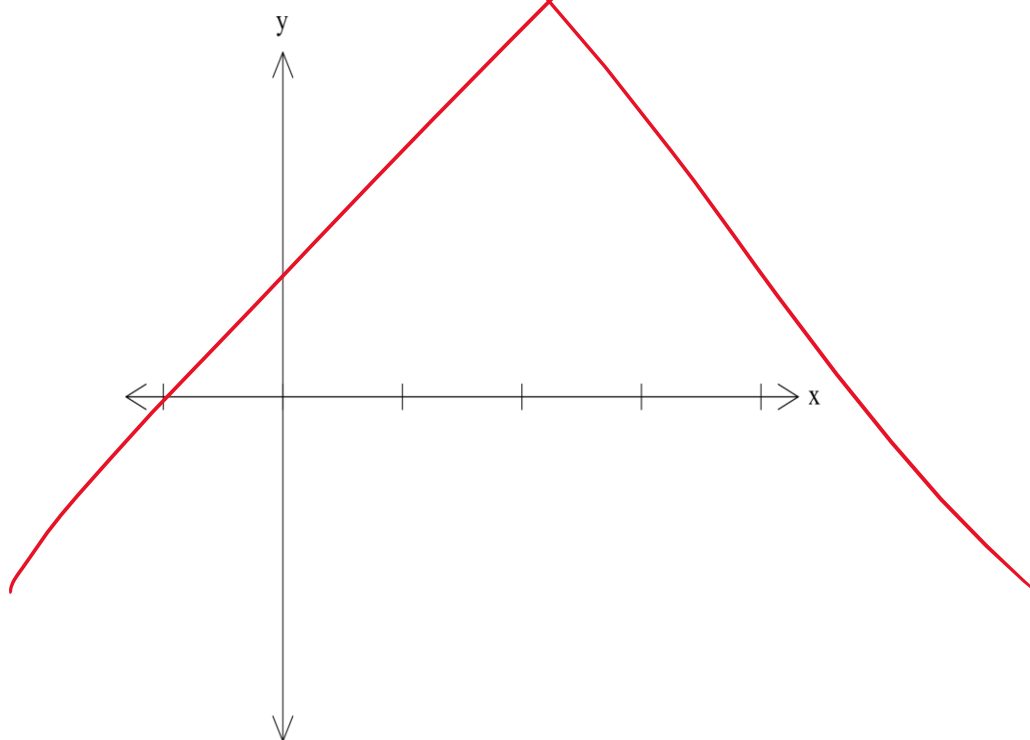
END OF TEST

Name

$$y = f(x)$$



$$y = f'(x)$$



Y11 3u 2015 prelim.

1a. (A)

b. $F'(x) > 0$ increasing (C).
 $F''(x) > 0$ concave up

c. $\Delta = (4)^2 - 4(1)(-3)$
 $= 16 + 12$
 $= 28 > 0$ (A).

d. $4x^2 + 3x - 5 \equiv Ax^2 + Ax + Bx + B + C$
 $4 = A \quad | \quad 3 = A + B$
 $3 = 4 + B$
 $-1 = B$ (B)

e. $A(4, -3) \quad B(-8, 5)$
 $3: -1$

$$x = \frac{-4 - 24}{a}, \quad y = \frac{3 + 15}{a}$$

$$x = -14, \quad y = 9.$$

(D).

#6a. $\frac{4}{x+2} \geq \frac{1}{x}$

consider $\frac{4}{x+2} = \frac{1}{x}$

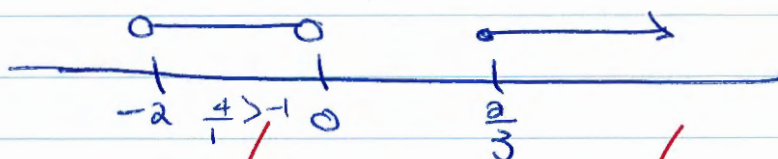
$x \neq 0$

$x \neq -2$

$4x = x + 2$

$3x = 2$

$x = \frac{2}{3}$ ✓



$-2 < x < 0$

$x \geq \frac{2}{3}$ ✓

3.

b. $x^2 - mx + 2 = 0$

roots: $\alpha, 2\alpha$

sum: $\alpha + 2\alpha = -\frac{b}{a}$

$3\alpha = m$ — (2)

prod: $(\alpha)(2\alpha) = \frac{c}{a}$
 $2\alpha^2 = 2$

$\alpha^2 = 1$ ✓

$\alpha = \pm 1$ — (1)

subst (1) in (2).

$3(\pm 1) = m$ ✓

$m = \pm 3$ ✓

→

c. $x = 3 - t$

$t = 3 - x$ — (1)

$y = 2t + 1$ — (2)

subst (2) in (1).

$y = 2(3 - x) + 1$ ✓

$y = 6 - 2x + 1$ ✓

$y = 7 - 2x$ ✓

→

$$d). 2x^2 - 13x + 9 = 0$$

$$\alpha + \beta = -\frac{b}{a} = \frac{13}{2}$$

$$\alpha\beta = \frac{c}{a} = \frac{9}{2}$$

$$\begin{aligned} i). (\alpha+1)(\beta+1) \\ &= \alpha\beta + \alpha + \beta + 1 \\ &= \frac{9}{2} + \frac{13}{2} + 1 \\ &= \frac{12}{2} \end{aligned}$$

$$\begin{aligned} ii). \frac{\alpha}{\beta} + \frac{\beta}{\alpha} \\ &= \frac{\alpha^2 + \beta^2}{\alpha\beta} \\ &= \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta} \\ &= \frac{\left(\frac{13}{2}\right)^2 - 2\left(\frac{9}{2}\right)}{\frac{9}{2}} \\ &= \frac{133}{18} \end{aligned}$$

$$e). \tan^2 x - 3 \sec x + 3 = 0.$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\therefore \sec^2 x - 1 - 3 \sec x + 3 = 0.$$

$$\sec^2 x - 3 \sec x + 2 = 0$$

$$(\sec x - 2)(\sec x - 1) = 0$$

$$\begin{array}{l|l} \sec x = 2 & \sec x = 1 \\ \frac{1}{\cos x} = 2 & \cos x = 1 \end{array}$$

$$\cos x = \frac{1}{2}$$

$$\text{ref } \angle = 60^\circ$$

cos is +

1st | 4th.

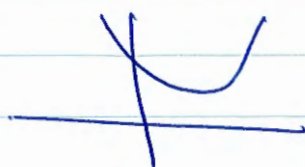
$$x = 60^\circ$$

$$x = 300^\circ$$

$$x = 0^\circ, 360^\circ$$

#7a. $Kx^2 - (5k+1)x + 9k$

for pos def



$a > 0$

$\Delta < 0$.

$k > 0$.

$\Delta = (- (5k+1))^2 - 4(k)(9k) < 0$ ✓
 $25k^2 + 10k + 1 - 36k^2 < 0$

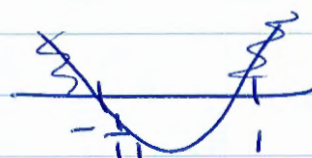
$-11k^2 + 10k + 1 < 0$

$11k^2 - 10k - 1 > 0$

$(11k+1)(k-1) > 0$

$k = -\frac{1}{11}, k = 1$ ✓

3.



and $k > 0$.

$\therefore k > 1$ ✓

b. $y^2 = 8x + 2y + 7$

$y^2 - 2y + (1)^2 = 8x + 7 + (1)^2$

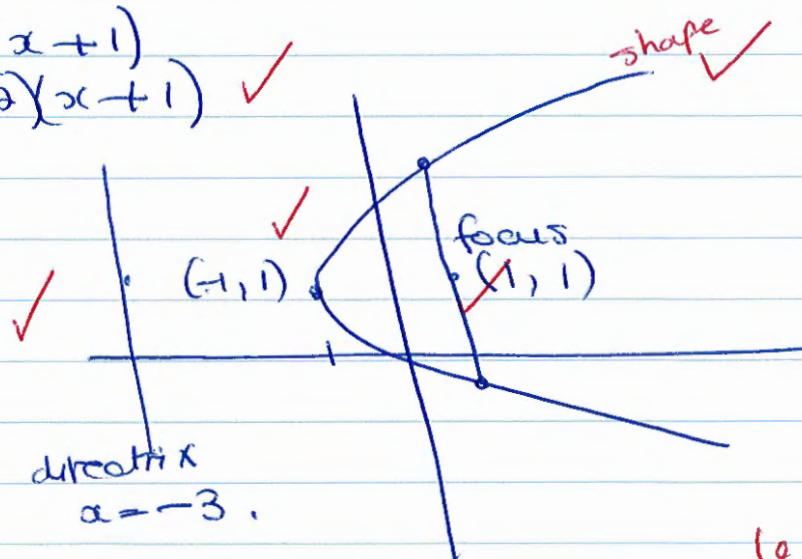
$(y-1)^2 = 8x + 8$

$(y-1)^2 = 8(x+1)$

$(y-1)^2 = 4(2)(x+1)$ ✓

✓ $(-1, 1)$

$a = 2$.



6

at $x = 1$

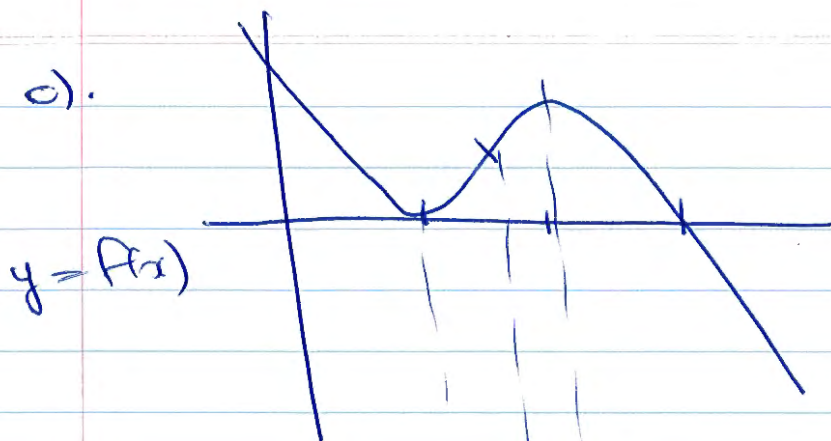
$(y-1)^2 = 16$

$y-1 = \pm 4$

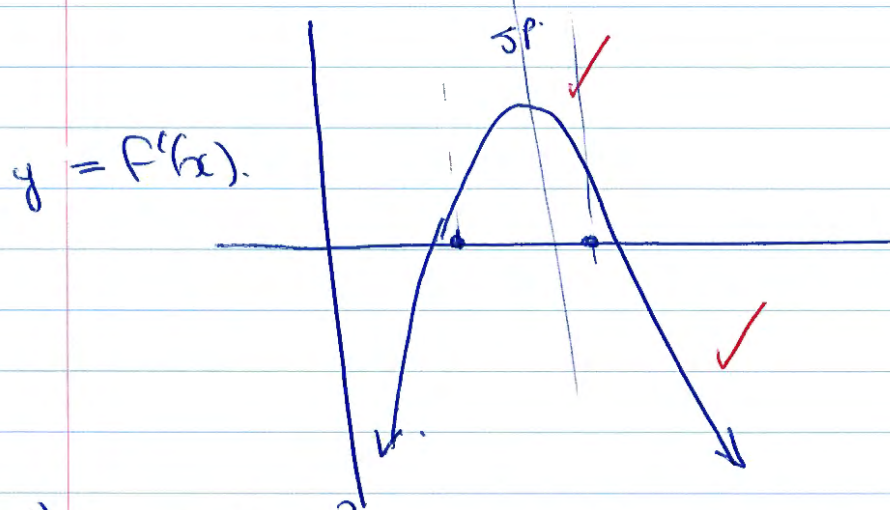
$y = 1 \pm 4$

$(1, 5)$ or $(1, -3)$ ✓

→



p.o.c.
↓
SP
↓
x.int.



2.

d).

$$y = \frac{x^2}{2a}$$

$$m(\text{tan}) = \frac{dy}{dx} = \frac{2x}{2a} = \frac{x}{a}$$

at $x = 2ap$.

$$m(\text{tan}) = \frac{2ap}{2a} = p$$

eq. of tangent: ✓

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2 \quad \text{--- (1) ✓}$$

∴ eq. at Q: $y = qx - aq^2$ --- (2) ✓

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p - q) = a(p - q)(p + q)$$

$$x = a(p + q) \quad \text{--- (3) ✓}$$

subst (3) in (1)

$$y = ap(p + q) - ap^2$$

$$y = ap^2 + apq - ap^2 = apq. \quad (a(p + q), apq) \quad \text{--- ✓}$$

4.

#8a. $y = \frac{x^2}{1-x^2}$

S.P. $y' = 0$. $u = x^2 \quad \left| \quad v = 1 - x^2 \right.$
 $u' = 2x \quad \left| \quad v' = -2x \right.$

$$y' = \frac{2x(1-x^2) - x^2(-2x)}{(1-x^2)^2} = 0.$$

$$2x - 2x^3 + 2x^3 = 0. \quad 3.$$

$$2x = 0$$

$$x = 0.$$

$$y = 0. \quad (0, 0) \text{ min S.P.}$$

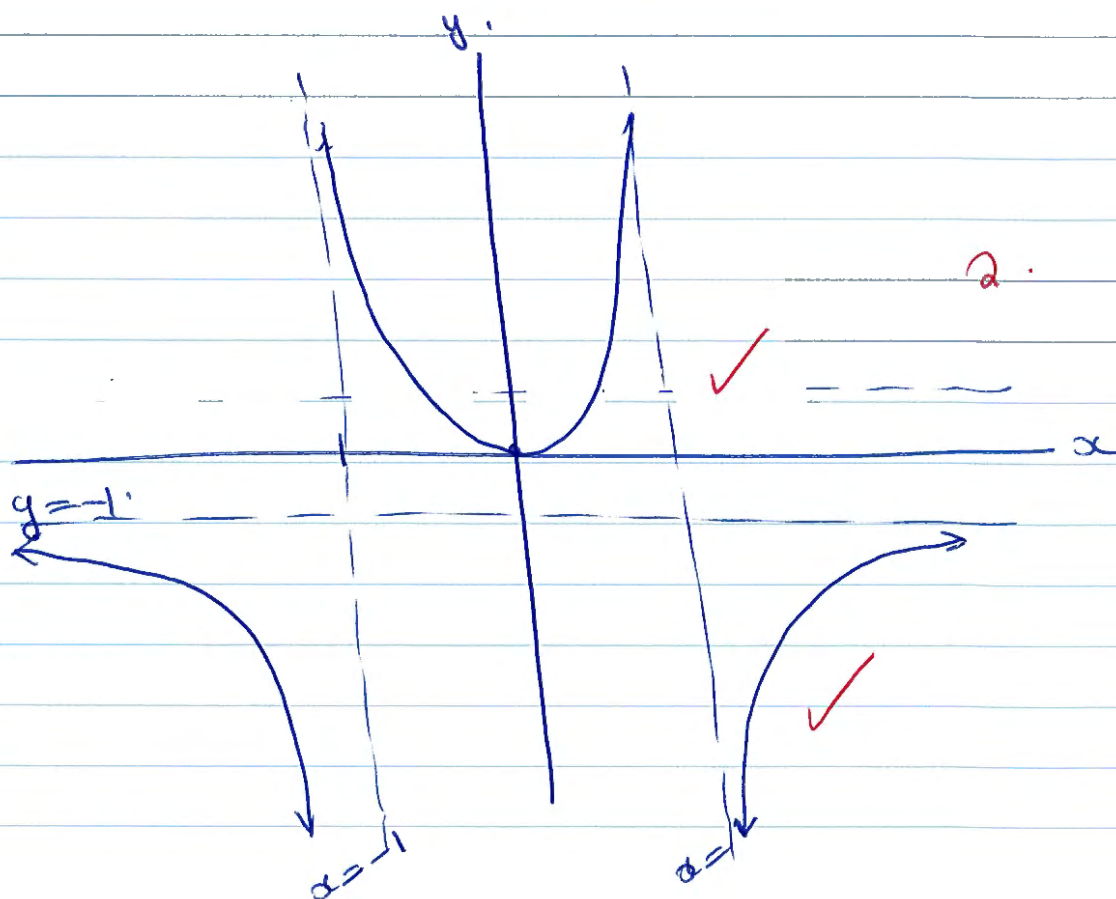
i).

x	$-\frac{1}{2}$	0	$\frac{1}{2}$
y'	$\backslash$	$\cup$	$/$

ii). $1 - x^2 \neq 0$.
 $x^2 + 1$
 $x \neq \pm 1.$

$$\lim_{x \rightarrow \infty} \frac{x^2/x^2}{\frac{1}{x^2} - x^2/x^2} = -1.$$

$$y = -1.$$



$$\#86. \quad 3 \times 2^2 + 3^2 \times 2^3 + 3^3 \times 2^4 + \dots + 3^n \times 2^{n+1} \\ = \frac{12}{5} (6^n - 1)$$

prove true for $n=1$.

$$\begin{array}{l|l} \text{LHS: } 3 \times 2^2 & \text{RHS} \\ = 12 & \frac{12}{5} (6^1 - 1) = \frac{12}{5} (5) = 12 \end{array}$$

$\therefore$ true for $n=1$.

assume true for $n=k$.

$$3 \times 2^2 + 3^2 \times 2^3 + \dots + 3^k \times 2^{k+1} = \frac{12}{5} (6^k - 1) \quad \checkmark$$

prove true for $n=k+1$.

we want to prove:

$$\underbrace{3 \times 2^2 + 3^2 \times 2^3 + \dots + 3^k \times 2^{k+1}}_{\text{by assumption}} + 3^{k+1} \times 2^{k+2} = \frac{12}{5} (6^{k+1} - 1) \quad \checkmark$$

LHS:

$$\frac{12}{5} (6^k - 1) + 3^{k+1} \times 2^{k+2} \quad \checkmark$$

$$= \frac{12}{5} (6^k - 1) + 3^k \cdot 3^1 \cdot 2^k \cdot 2^2$$

$$= \frac{12}{5} (6^k - 1) + 6^k \cdot 12$$

$$= \frac{12}{5} \cdot 6^k - \frac{12}{5} + 12 \cdot 6^k$$

$$= \frac{12 \cdot 6^k - 12 + 12 \cdot 5 \cdot 6^k}{5} \quad \checkmark$$

$$= \frac{12}{5} (6^k - 1 + 5 \cdot 6^k)$$

$$= \frac{12}{5} (6^1 \cdot 6^k - 1) \quad \checkmark$$

$$= \frac{12}{5} (6^{k+1} - 1) = \text{RHS} \quad \checkmark$$

4

$$8c). (x^2 + 2x)^2 = 14x^2 + 28x + 15$$

$$(x^2 + 2x)^2 = 14(x^2 + 2x) + 15$$

$$\text{let } x^2 + 2x = k \quad \checkmark$$

$$k^2 = 14k + 15$$

$$k^2 - 14k - 15 = 0$$

$$(k-15)(k+1) = 0$$

$$k = 15$$

$$k = -1 \quad \checkmark$$

$$x^2 + 2x = 15$$

$$x^2 + 2x = -1$$

$$x^2 + 2x - 15 = 0$$

$$x^2 + 2x + 1 = 0$$

$$(x+5)(x-3) = 0$$

$$(x+1)^2 = 0$$

$$x = -5, 3 \quad \checkmark$$

$$x = -1 \quad \checkmark$$

← 0.